Homework 4 - 09/17 Content

MATH 1271, Garrett Credi

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Problems listed as AC are found in the Active Calculus textbook at the end of the section listed.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed section.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Limits at Infinity

Choose 4 of the following problems to solve.

1. PAULS-ASSIGN.Limits.Limits at Infinity Part 1.4-17 (each of these counts as one problem)
2. APEX 1.6.19-28 (each of these counts as one problem)

2 Type 2 - Limits Diverging to Infinity

Choose 4 of the following problems to solve.

1. PAULS-ASSIGN.Limits.Infinte Limits.1-8 (each of these counts as one problem)
2. PAULS-ASSIGN.Limits.Infinte Limits.9-12 (each of these counts as one problem)

3 Type 3 - The Squeeze Theorem

Choose 2 of the following problems to solve.

1. APEX 1.3.35-38 (each of these counts as one problem)
2. PAULS-ASSIGN.Limits.Computing Limits.31-35 (each of these counts as one problem)

4 Challenge Problem

Recall the following facts/definitions:

1. The *degree* of a polynomial $p(x)$ is the highest power appearing in the polynomial, and it is denoted by $\deg(p(x))$. E.g. $\deg(x^2) = 2, \deg(x^5 + 4x^2 + 2x + 1) = 5$, etc.
2. The *leading term* of a polynomial $p(x)$ is the monomial summand with the highest degree, and it is denoted by $LT(p(x))$. E.g. $LT(x^3 + 3x^2 + 2x + 1) = x^3, LT(10x^5 + 3x^2 + 1) = 10x^5$ etc.
3. The *leading coefficient* of a polynomial $p(x)$ is the coefficient of the leading term, and it is denoted by $LC(p(x))$.
4. A *rational function* is a function that can be expressed as the ratio of two polynomials.

Show that the following facts about limits at infinity of a rational function

$$\frac{p(x)}{q(x)} = \frac{a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + ... + b_1 x + b_0} = \frac{\sum_{i=0}^{n} a_i x^i}{\sum_{j=0}^{m} b_j x^j}$$

hold:

1. If $\deg(p(x)) > \deg(q(x))$ then:
   (a) If $\deg(p(x))$ is an *even* number,

$$\lim_{x\to\infty} \frac{p(x)}{q(x)} = \lim_{x\to-\infty} \frac{p(x)}{q(x)} = \begin{cases} \infty, & LC(p(x)) > 0 \\ -\infty, & LC(p(x)) < 0 \end{cases} \quad (1)$$

(b) If $\deg(p(x))$ is an *odd* number,

$$\lim_{x\to\infty}\frac{p(x)}{q(x)} = -\lim_{x\to-\infty}\frac{p(x)}{q(x)} = \begin{cases}\infty, & LC(p(x)) > 0\\ -\infty, & LC(p(x)) < 0\end{cases} \tag{2}$$

2. If $\deg(p(x)) = \deg(q(x))$ then

$$\lim_{x\to\infty}\frac{p(x)}{q(x)} = \lim_{x\to-\infty}\frac{p(x)}{q(x)} = \frac{LC(p(x))}{LC(q(x))} \tag{3}$$

3. If $\deg(p(x)) < \deg(q(x))$ then

$$\lim_{x\to\infty}\frac{p(x)}{q(x)} = \lim_{x\to-\infty}\frac{p(x)}{q(x)} = 0 \tag{4}$$